vitamin D

* its main function is the regulation of calcium absorption and Homeostasis
* Synthesized in the Skin
* normal ditery form (active form) of vitamin D is cholecalciferol (calciol)

Types :

① VD_2 (Ergocalciferol) ② VD_3 (cholecalciferol)

active form of ViD_3 // $1,25(OH)_2$-ViD_3

trasportaition // DBP → (vitamin D binding Protein) (transcalciferin)

Absorption and metabolism :

Vitamin D is absorbed in lipid micelles and incorporated into chylomicrons.

Synthesis of Vitamin D :

- pre-pro-ViD_3 (7-dehydrocholesterol)
- pro-ViD_3 (7-hydrocholesterol)
- active form ViD_3 (cholecalciferol)

(undergous non-enzymatic reaction on exposure to UV light)

Synthesis of cholecalciferol is High in Summer

Synthesis of cholecalciferol is low in winter

functions of Vitamin D :

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the principal function of Vitamin D is to maintain the plasma concentration of Calcium in this ways :

1. increased intestinal absorption of Calcium
2. reduced excrition of Calcium by Stimulating reabsorption
3. mobilization of bone minral
4. Secretion and Synthesis of insulin, thyroid, Parathyroid hormones.
5. inhibition of Production of interleukin, immunoglobulin
6. modulation of cell diffrentiation, proliferation, apoptosis

Rickets :

their bones are undermineralized as a result of poor absorption of calcium in the absense of inadequate ammount of Calcitriol.

Clinical features of rickets :

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1. wide Sutures
2. rachitic rosary
3. Bowing of legs
4. Dental Hypoplasia
5. Swelling in wrist and ankle joints.
6. frontal bossing

Osteomalacia :

it result from deminerlization of bones, rather than the failure to minerlize it in the first place.

* iron defieeiny decrese in Vitamin absorption *

* vitamin D metabolism in the kidney is regulated *

* ditery Vitamin D is absorbed in intestinal cells *

mcq ← * 1.50 of vitamin D is absorbed *

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